Journal of ELECTRONIC MATERIALS
https://doi.org/10.1007/s11664-020-07976-4
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A First-Principles Study of the Adsorption of H_2O on Ru- and Mo-Alloyed Pt(111) Surfaces

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A study on the molecular adsorption of a water (H_2O) monomer on Pt alloy surfaces with binary Pt-Ru and Pt-Mo and ternary Pt-Ru-Mo surface models was conducted. Some calculations of the interaction between a H_2O monomer and the Pt surface were also done for reference. This study is based on density functional theory (DFT) with periodic models and aims to understand the H_2O adsorption mechanism. In Pt, Pt-Ru, and Pt-Mo surfaces, H_2O preferably adsorbs via an oxygen atom in a flat configuration at the top sites of Pt, Ru, Mo, respectively. However, on the ternary Pt-Ru-Mo surface, the adsorption configuration has the most inclined H_2O structure relative to the surface. The results showed that the binding energy of H_2O/Pt-Ru-Mo > H_2O/Pt-Mo > H_2O/Pt-Ru > H_2O/Pt. The adsorption mechanism was then clarified by charge transfer and natural bonding. The charge transfer from the surface to the adsorbate is observed in all models, with the greatest charge transfer occurring on the surface of Pt doped with two Mo atoms. This is probably due to the fact that oxygen can attract the most charge on Mo, because the difference in electronegativity is greatest. The calculation results also show that Ru is the most hydrophilic metal for oxygen. However, since the adsorption structure is parallel to the surface, hydrogen (H) is also more sensitive to receive the charge. Subsequently, the most acceptable reason for the most stable adsorption for H_2O/Pt-Ru-Mo is that the inclined structure yields the most orbitals hybridization at the H_2O's highest occupied molecular orbital (HOMO). This drives the interaction by forming bonding states at the lowest energy and antibonding states at the highest energy.

Key words: H_2O, Pt alloy surfaces, DFT, charge transfer, adsorption mechanism

INTRODUCTION

The need to transfer from oil energy sources to hydrogen energy sources is urgently needed because of the reduced availability of fuel oil. This is because oil exploitation is continuous, while the production of fuel oil from organic materials takes a long time. In addition, environmental problems caused by fuel oil consumption should be taken seriously. These facts encourage scientists and engineers around the world to develop better energy storage technologies and power converters to meet global energy needs. Both devices must be optimal in terms of the availability of abundant raw materials, easy to synthesize, cheap, and environmentally friendly to operate. One answer to this problem is a direct methanol fuel cell (DMFC), a device that converts methanol into energy through an electrochemical process. In this device, methanol solution acts as fuel in DMFCs that are oxidized at the anode during

(Received June 22, 2019; accepted January 25, 2020)

Published online: 10 February 2020

their operation. The oxidation process is usually catalyzed by metal surfaces. The generally accepted oxidation processes are the dehydrogenation of methanol by intermediate reactions and the decomposition of water molecules as oxidants.[1–3] Since the anode DMFCs contain methanol dissolved in water, the interaction between the methanol molecule and the catalyst metal surface as well as the interaction between the water molecules and the catalyst metal surfaces becomes a necessity. Thus, one of the fundamental interactions that occurs in the conversion of methanol into energy is the interaction between water molecules and metal catalysts at the anode. The anode materials that interact with water molecules affect the overall reactions and further determine the performance of the device. That is why researchers are still working on adjusting and constructing the material with the goal of achieving optimal technological products.

One of the material designs for optimizing the oxidation reactions at the anode of DMFCs is the alloying of Pt with Ru,[4] Ni,[5] Co,[6] Mo,[7] and Fe.[8] In addition, a further improvement in the use of a ternary alloy is used.[5,8] Most of these and other papers explain the increase in cell performance due to the addition of second and third metals through the ligand effect and the bifunctional mechanism associated with the poisoned species of methanol degeneration.[9–12] Therefore, we are investigating a more fundamental aspect of adsorption of water molecules on Pt alloy surfaces to understand and clarify the adsorption mechanism. The investigation of the fundamental interaction of such a system could be extended to other electrochemical reactions such as promoter systems and catalysts,[13] corrosion systems,[14] and so on. In addition, the study of water on the surface of Pt(111) is realistic because many ultrahigh vacuum (UHV) experiments confirm their existence with various techniques, e.g., spectral scattering and temperature-programmed desorption (TPD) measurements[15] and calorimetric measurements.[16] Based on the importance of the interaction between water and metal surfaces, especially Pt, and based on experimental facts about their existence in the physical system, this investigation aimed to provide a qualitative picture of H_2O adsorption on the Pt alloy surfaces.

The interaction of H_2O and a Pt(111) surface is extremely difficult to characterize experimentally, since their interaction is comparable to the H_2O–H_2O interaction. It is said that the formation of water clusters takes place rapidly at a temperature of 60 K.[15] This obscures the understanding of the natural H_2O–Pt interaction. As time progressed, there was good news when the advanced scanning tunneling microscope technology with action spectroscopy (STM-AS) was able to detect monomers and dimers on the Pt(111) surface at a very low coverage.[17] Using STM-AS, it is observed that water monomer's center of mass is imaged at the top site. Since this work deals with Pt(111) surfaces alloyed with Ru and Mo at low coverage, we used a water monomer model to clarify the water–surface interaction. To achieve this goal, we used density functional theory (DFT) to calculate the H_2O adsorption energy on each surface and then analyze the charge transfer that occurs during the adsorption process. DFT investigations on the adsorption of H_2O on metal surfaces have been carried out; for example, pure Pt surfaces with different Miller indices,[18,19] various pure surfaces of rare earth metals,[20] and systems of bimetal alloys[21,22]. Thus, here, we used binary Pt-Ru and Pt-Mo and ternary Pt-Ru-Mo as surface models. As far as our literature review is concerned, no detailed reports on the water adsorption on these alloy surfaces have been found in connection with the effects of additional metals and their analysis of the adsorption mechanism.

COMPUTATIONAL DETAILS

We investigated the molecular water adsorption on Pt-Ru, Pt-Mo, and Pt-Ru-Mo (111) surfaces based on DFT calculations employing the Vienna ab initio simulation package (VASP).[23,24] All calculations are performed using the projector augmented-wave (PAW) method,[25] i.e., by replacing the strong interaction of nuclei and electrons with ultrasoft pseudopotentials and by expanding the Kohn–Sham orbitals in a plane-wave basis set. Therefore, in this calculation, we adjusted the size of the plane-wave basis set with the cutoff energy of 400 eV because of the stronger oxygen core. In addition, we used the Perdew and Wang (PW91) version of the generalized gradient approximation (GGA)[26,27] to guarantee the computational accuracy due to the electron exchange correlation effect. This choice is based on the fact that the binding parameters of platinum and water, which are reproduced from GGA, were found to be reasonable in other calculations, and the PW91 gave excellent results in the description of water intermolecular interactions.[28] Integration across the Brillouin zone in reciprocal space was calculated numerically by the specific $5 \times 5 \times 1$ k-point sampling of the Monkhorst–Pack scheme.[29] Then, for relaxation calculations, we used order one of the Methfessel–Paxton method with smearing of 0.2 eV.[30] The electronic self-consistent iteration is set at 10^{-4} eV for its convergence criteria, while ion relaxation is converged when the Hellman–Feyman forces drop to less than 0.001 eV/Å on atoms. In addition, we also calculated the dipole corrections to anticipate the artificially generated electric field across the slab.[31]

The surfaces of Pt-Ru, Pt-Mo, and Pt-Ru-Mo (111) were constructed by displacing the atomic Pt with Ru, Mo, and both Ru and Mo into the pure Pt(111) surface. As the host material, a Pt(111) surface was obtained by cutting bulk platinum and taking the experimental lattice constant value of 3.92 Å.[32] Our surfaces of face-centered cubic (111) were made in a

supercell model with a three-layer slab, and each layer contains 16 atoms (a 4 × 4-unit cell). Each slab was separated by a sufficiently large vacuum of ∼15 Å to prevent interaction between H_2O and the next slab. The atomic positions of the top two layers were relaxed, while all the atoms in the bottom layer were kept constant at the bulk positions. Small numbers of atomic Ru in Pt-Ru, atomic Mo in Pt-Mo, and Ru and Mo in Pt-Ru-Mo mimic the experiments.[33,34] Small amounts of Ru and Mo dopants in our surface models are represented by substitution of two Ru atoms or two Mo atoms to the surface of Pt(111) consisting of 48 atoms. In this way, we proposed Pt-Ru and Pt-Mo surface models, while the Pt-Ru-Mo surface model is obtained by replacing a Ru atom and a Mo atom on the same Pt(111) surface. The difference between the total energies of the alloy surfaces of Pt-Ru, Pt-Mo, Pt-Ru-Mo, and the pure Pt surface, i.e., the negative surface segregation energy of the pure Pt, means the platinum (Pt) as host material in the surface systems. With regard to the surface stability of Pt-Ru-Mo in our previous study,[35] we therefore used surfaces of Pt-Ru, Pt-Mo, and Pt-Ru-Mo to investigate the influence of Ru and Mo on the water adsorption mechanism. For completeness and at the same time as a reference, we also briefly show the results of important calculations on the interaction of H_2O with the pure Pt(111) surface. The surface models in this study and all possible adsorption sites for the adsorption of H_2O on Pt, Pt-Ru, Pt-Mo, and Pt-Ru-Mo surfaces are then depicted in Fig. 1(a.i–d.i). Here, we calculated the potential energy of such a surface-adsorbate interaction at all sites shown in Fig. 1 (left, labeled **i**) to determine the most preferred adsorption configuration. Here, the most stable structure is defined in terms of adsorption energy per molecule, i.e.,

$$E_{\mathrm{ads}} = \left[E_{(\mathrm{H_2O})_n/\mathrm{surf}} - (E_{\mathrm{surf}} + nE_{\mathrm{H_2O}})\right]/n, \quad (1)$$

where the most likely adsorption occurs when E_{ads} has the most negative energy. E_{ads} denotes the adsorption energy of H_2O on associated surfaces, $E_{(\mathrm{H_2O})_n/\mathrm{surf}}$ indicates the total energy of the entire system, E_{surf} represents the total energy of clean surfaces, $E_{\mathrm{H_2O}}$ is the total energy of isolated H_2O, and n is the number of water molecules. In this study, we only used models of a water dimer with n = 2 for the first comparison with experimental results of the adsorption structure of a single water dimer on Pt(111) and a water monomer with n = 1 for the whole model being investigated. The calculated total energies obtained from Eq. (1) are then corrected due to the difference between the zero-point energy (ΔE_{ZPE}) of H_2O in adsorbed phase (H_2O, ads) and H_2O in isolated phase (H_2O, iso) which is expressed by

$$\Delta E_{\mathrm{ZPE}} = \left(\sum_i \frac{\hbar\omega_i}{2}\right)_{\mathrm{H_2O,ads}} - \left(\sum_i \frac{\hbar\omega_i}{2}\right)_{\mathrm{H_2O,iso}} \quad (2)$$

where $\hbar$ is Plank's constant and ω is the vibrational frequencies of the H_2O molecules in the adsorbed (ads) and isolated phase (iso).

Furthermore, the water adsorption mechanism was evaluated by charge transfer occurring in the adsorption process. The charge transfer was verified by analyzing the charge density difference which is calculated by

$$\Delta n(\vec{r}) = n_{\mathrm{H_2O/surf}}(\vec{r}) - (n_{\mathrm{surf}}(\vec{r}) + n_{\mathrm{H_2O}}(\vec{r})), \quad (3)$$

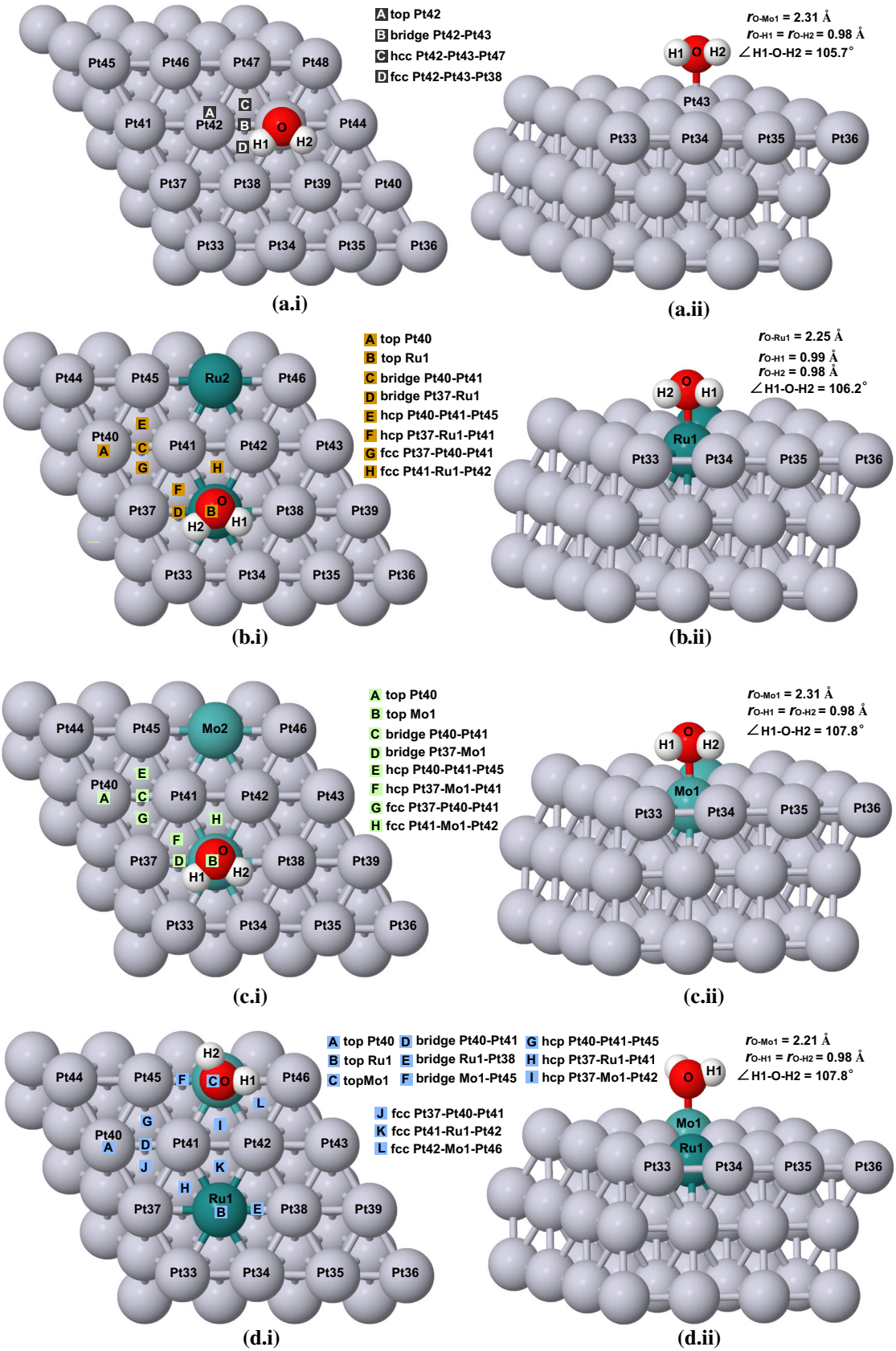
where $n_{\mathrm{H_2O/surf}}(\vec{r})$, $n_{\mathrm{surf}}(\vec{r})$, and $n_{\mathrm{H_2O}}(\vec{r})$ are the charge density of H_2O adsorbed on surfaces, surfaces, and water molecules, respectively. In addition, analysis of the electronic structure and binding properties formed between the surface and the adsorbate was performed using the density of states (DOS).

RESULTS AND DISCUSSION

As it is known, a water molecule adsorbs on the metal surfaces in an upright or downward configuration. The upright configuration refers to the position of the two H atoms of H_2O that are located above the O atom, and conversely, the down configuration. Between them is a flat configuration in which all the atoms in H_2O are parallel to the surface. In our surface models, the most stable configurations for adsorption of water monomers are parallel for binary Pt-Ru and Pt-Mo and nearly parallel for ternary Pt-Ru-Mo with respect to the surfaces. This flat configuration of the H_2O molecule in the gas phase gives a gap between the highest occupied molecular orbitals (HOMOs) and the lowest unoccupied molecular orbitals (LUMOs) of 7.6 eV. This calculated HOMO-LUMO gap is in the experimental range of Grand et al.[36] with a calculated gap of 7.0 eV and Goulet et al.[37] with a width of 9.0 eV. The HOMO of a parallel configuration is the $1b_1$-O "lone pair" orbital. This $1b_1$ orbital in H_2O molecules is very close to the Fermi level. Moreover, to ensure that the calculations made are justified, we also calculated the vibrational frequency of H_2O molecules in the gas phase. The three frequencies we observed were 3816 cm^{-1}, 3682 cm^{-1}, and 1554 cm^{-1}, respectively, for the vibration modes of asymmetrical stretching, symmetrical stretching, and bending deformation. These values are in excellent agreement with the experimental data of Sexton (1980).[38]

INTERACTION OF H_2O MOLECULES WITH Pt(111)

The interaction of water molecules with Pt(111) surfaces has been studied for many years. Here, we present brief results of important features to check


A top Pt42
B bridge Pt42-Pt43
C hcc Pt42-Pt43-Pt47
D fcc Pt42-Pt43-Pt38
(a.i)
$r_{O\text{-}Mo1}$ = 2.31 Å
$r_{O\text{-}H1}$ = $r_{O\text{-}H2}$ = 0.98 Å
∠H1-O-H2 = 105.7°
(a.ii)
A top Pt40
B top Ru1
C bridge Pt40-Pt41
D bridge Pt37-Ru1
E hcp Pt40-Pt41-Pt45
F hcp Pt37-Ru1-Pt41
G fcc Pt37-Pt40-Pt41
H fcc Pt41-Ru1-Pt42
(b.i)
$r_{O\text{-}Ru1}$ = 2.25 Å
$r_{O\text{-}H1}$ = 0.99 Å
$r_{O\text{-}H2}$ = 0.98 Å
∠H1-O-H2 = 106.2°
(b.ii)
A top Pt40
B top Mo1
C bridge Pt40-Pt41
D bridge Pt37-Mo1
E hcp Pt40-Pt41-Pt45
F hcp Pt37-Mo1-Pt41
G fcc Pt37-Pt40-Pt41
H fcc Pt41-Mo1-Pt42
(c.i)
$r_{O\text{-}Mo1}$ = 2.31 Å
$r_{O\text{-}H1}$ = $r_{O\text{-}H2}$ = 0.98 Å
∠H1-O-H2 = 107.8°
(c.ii)
A top Pt40
B top Ru1
C topMo1
D bridge Pt40-Pt41
E bridge Ru1-Pt38
F bridge Mo1-Pt45
G hcp Pt40-Pt41-Pt45
H hcp Pt37-Ru1-Pt41
I hcp Pt37-Mo1-Pt42
J fcc Pt37-Pt40-Pt41
K fcc Pt41-Ru1-Pt42
L fcc Pt42-Mo1-Pt46
(d.i)
$r_{O\text{-}Mo1}$ = 2.21 Å
$r_{O\text{-}H1}$ = $r_{O\text{-}H2}$ = 0.98 Å
∠H1-O-H2 = 107.8°
(d.ii)

◀Fig. 1. (Left, labeled i) All possible adsorption sites of (a.i) Pt, (b.i) Pt-Ru, (c.i) Pt-Mo, and (d.i) Pt-Ru-Mo surfaces. The adsorption energy of a H_2O molecule at all these sites is calculated. (Right, labeled ii) The most stable adsorption geometry of a water molecule on (a.ii) Pt, (b.ii) Pt-Ru, (c.ii) Pt-Mo, and (d.ii) Pt-Ru-Mo surfaces

the experimental compatibility of the model. The importance of this step is to ensure that our model is realistic and accountable. First, we will report the H_2O dimer adsorption structure along with adsorption energy. This is based on the success of the experiment in displaying the STM image of the H_2O dimer/Pt(111) and in determining the vibrational spectra.[17] Our calculation of the geometric structure of the water dimer/Pt(111) agrees very well with the study of the STM image and the DFT by Motobayashi et al.[17] The associated adsorption energy that we obtained was 0.41 eV. The plot of charge profile also indicates that no charges accumulate between two H_2O molecules. This suggests the possibility of observing the H_2O monomer.

Next, we will also report on H_2O monomer adsorption on Pt(111), since the STM image for H_2O/Pt(111) monomers can also be predicted for the very low H_2O coverage, which is less than 0.005 ML.[17] On the other hand, the infrared spectra of the H_2O monomer on Ru(0001) were successfully observed.[39] These facts convince us that the modeling of H_2O monomer adsorption over Pt(111) is reasonable. The interaction between a H_2O monomer and the Pt(111) surface is characterized by an absorption energy of ~0.30 eV. This result is very close to the DFT calculation by Fajin et al.[18] and Meng et al.[20] and is a correction of our previous result.[40] Figure 1(a.ii) shows that H_2O adsorbs on Pt(111) via the O atom of H_2O with the adsorption site at the top. This result confirms the prediction of the STM image by Motobayashi et al.[17] The H_2O configuration during adsorption runs parallel to the Pt(111) surface, with the H1–O–H2 angle widening to 105.7°. The other important parameter that can be measured experimentally and that is available in the adsorption system is the vibrational frequency. The vibrational modes that we calculated were asymmetrical stretching, symmetrical stretching, and bending deformation of H1–O–H2 and O–Pt stretching of 3683 cm^{-1}, 3583 cm^{-1}, 1556 cm^{-1}, and 189 cm^{-1}, respectively. These first three modes are close to the vibration frequencies measured by Sexton.[38] He obtained 3400 cm^{-1}, 3400 cm^{-1}, and 1625 cm^{-1} for those three types of H1–O–H2 vibration modes. However, our calculated O–Pt stretching is significantly underestimated by the experimentally measured ones. Sexton measured it and obtained 550 cm^{-1} for O–Pt stretching. Overall, all modes observed in our calculations are smaller than those recorded with high-resolution electron energy loss spectra (HREELS). It also occurs when our DFT calculations of monomer H_2O/Pt(111) are compared with observations of vibration signals of dimer H_2O/Pt(111) with STM-AS.[17] This can be understood by the difference between the adsorbate used as the object. Both experiments used a water cluster as a sample, whereas this model is a H_2O monomer.

Finally, we report the charge redistribution on the adsorbate and surface that occurs due to the interaction between them. This calculation is based on the Bader analysis.[41] All calculations in this article use a script developed by the group of Henkelman[42] taking into account the vacuum error corrections. The charge of atomic O and Pt in contact redistributes and becomes minimal due to the adsorption. This is a natural place to separate atoms from each other. Including the error bars from the H_2O vacuum charge of 0.02 e^-, the Pt surface of 0.24 e^-, and 0.25 e^- of H_2O/Pt, we observed that the water molecule received additional charge of 0.21 e^-. Thus, we conclude that the charge transfer is observed from substrate to adsorbate.

H_2O ADSORPTION ON THE Ru-ALLOYED Pt(111)SURFACE

On the surfaces of binary Pt-Ru, H_2O preferably adsorbs in a parallel configuration. The geometrical structure of this system is described in Fig. 1(b.ii). The water molecule is adsorbed on the surface by an O atom with a Ru–O bond length of 2.25 Å. The adsorption energy obtained in this case is 1.04 eV. In comparison to the previous result, it can be seen that the substitution of atomic Ru for Pt stabilizes the adsorption of H_2O relative to the adsorption on pure Pt(111). The angle between H1, O, H2 (∠H1–O–H2) is 106.2°, which is wider than the gas phase angle of 104.6°. Further calculation of the vibration frequency of the water molecule adsorbed on the Pt-Ru surface revealed that the asymmetrical stretching mode is 3590 cm^{-1}, the symmetrical stretching mode is 3432 cm^{-1}, and the bending deformation mode is 1485 cm^{-1}. These values are lower than the vibrational frequency of the water molecule in the gas phase. Thus, from the information about the value of the H_2O–surface interaction strength, the widening of the H_2O angle during the adsorption process, and the calculated H_2O vibration frequency, one may suggest that the H_2O can dissociate at low energy cost on the Pt-Ru surface. Finally, we also calculated the charge redistribution of the Pt-Ru surface due to the interactions with H_2O based on the Bader charge analysis.[41] Here, the Ru and O atoms, which interact directly with each other, seem to reach the lowest charge among the atoms. We also found that after adsorption, H_2O received 0.35 e^- of charge. This verifies whether the charge is transferred from the surface to the water molecule. Here, the water molecule gained more charge compared to the charge transfer calculated in H_2O/Pt(111).

H_2O ADSORPTION ON THE Mo-ALLOYED Pt(111) SURFACE

In the case of adsorption of H_2O on the Pt-Mo surface, the interaction strength between surface and adsorbate is higher than that of Pt-Ru. The calculated adsorption energy is 1.37 eV. This energy is required to bind the H_2O molecule at the top of Mo with a flat configuration, as shown in Fig. 1(c.ii). Here, the Mo and O atoms are bonded to each other with a bond length of 2.21 Å. The shorter bond length compared to Ru-O could be related to the adsorption energy of H_2O/Pt-Mo, which is higher to that of H_2O/Pt-Ru. In this system, the angle of H1–O–H2 of 107.8° is also greater than H_2O/Pt-Ru. Due to the higher adsorption energy and the larger H_2O angular expansion, we can assume that H_2O can be split more easily in Pt-Mo than in Pt-Ru. On the other hand, the calculated vibration frequencies for adsorbed H_2O on Pt-Mo in all cases are smaller than in isolated H_2O. We obtained the value of 3652 cm^{-1} for symmetrical stretching, 3552 cm^{-1} for symmetrical stretching, and 1445 cm^{-1} for bending-deformation modes. It may be related to the bond weakening between O and H in the H_2O molecule. Finally, we also calculated the atomic charge difference distribution before and after adsorption. The Mo atom, which interacts directly with H_2O, releases the most electrons compared to other neighboring atoms. In the adsorbate, the O atom also minimizes the charge density during adsorption. In the case of the monomer H_2O/Pt-Mo, we also verify the charge transfer of 0.40 e^- from the substrate to the adsorbate. The calculated charge transfer in H_2O/Pt-Mo is the highest among other models. This could be explained by considering electronegativity, i.e., the ability of atoms to attract electrons in their chemical bonds. Mo has a lower electronegativity compared to Ru, and, therefore, Mo does not as readily attract electrons from the O atom of H_2O. In contrast, atomic O with higher electronegativity can easily attract Mo electrons. In this case, we also observed the charge transfer from the water to the adsorbent upon adsorption. Moreover, we have also observed that the H_2O adsorption process in Pt-Mo involves physisorption and chemisorption. The physisorption begins when the H_2O molecule approaches the surface, which then undergoes chemisorption with an adsorption energy of 0.56 eV at a distance of 2.70 Å from the surface. This transition requires an activation energy of 0.004 eV.

H_2O ADSORPTION ON THE Ru- AND Mo-ALLOYED Pt(111) SURFACE

Finally, we come to the surface model of ternary Pt-Ru-Mo, which was constructed by replacing a Mo atom and a Ru atom on the Pt(111) surface. The use of these three surface models is intended to determine the effect of each additional metal on water adsorption. Interestingly, unlike other cases, the water molecule on the surface of Pt-Ru-Mo has a nearly flat configuration, with OH tilted upward to an angle of ∼15° to the surface, as shown in Fig. 1(d.ii). In this case, H_2O adsorbs preferentially at the top of the Mo atom. The adsorption energy of H_2O at this adsorption site is 1.50 eV. As observed on the Pt-Mo surface, the molecular adsorption of H_2O includes physisorption and chemisorption. The transition occurs at z = 2.75 Å and is activated by 0.058 eV. At z = 2.21 Å exactly on top of the Mo atom, the H_2O molecule reaches the most stable position with an OH elongation of 0.02 Å relative to the H_2O molecule in the gas phase and the angle of H1–O–H2 of 107.8°. The OH elongation and the H1–O–H2 angle widening may also indicate the tendency for dissociative adsorption. In addition, the calculated vibrational frequencies for water adsorbed on Pt-Ru-Mo are 3634 cm^{-1}, 3522 cm^{-1}, and 1528 cm^{-1} for asymmetrical stretching, symmetrical stretching, and bending-deformation modes, respectively. Compared to the vibrational frequency modes of the water molecule in the gas phase, these values are also smaller, which may indicate a weakening of the bond between O and H in the H_2O molecule after adsorption. The O–H bond weakening is then proposed for H_2O cleavage. In addition, the redistribution of the atomic charge due to adsorption is also calculated. The same occurrence of charge minimization of the atoms that interact directly, i.e., Mo and O atoms, is also observed in this system. The total charge of the water molecule achieved during the adsorption process is 0.28 e^-. Again, we confirm the charge transfer from the substrate to the adsorbate.

From the cases of adsorption of a H_2O molecule on the three surface models, we see the difference in adsorption geometry and adsorption energy. It can be assumed that these different adsorption configurations influence different electronic properties. It is therefore important to analyze the plausible reason for explaining the stronger binding in terms of the difference in a H_2O adsorption configuration. The following section gives the electronic properties of the three systems.

ELECTRONIC PROPERTIES

In order to understand the interaction between adsorbates and surfaces, we analyze the density of states (DOS) and the partial charge density, as shown in Fig. 2. The DOS used to interpret the chemical bonding properties between metals and H_2O is the partial density of states (PDOS) of s, p_y, p_z, and p_x orbitals of H_2O and d orbitals of the metal surface, which play the most important role in the interaction with H_2O. Partial charge densities serve to clarify the charge profile at a certain energy level and to characterize the formation of bonds during adsorption. Figure 2(a.i–d.i) and (a.ii–d.ii) shows the PDOS of the H_2O and contacting metals immediately before and after the interaction, while the

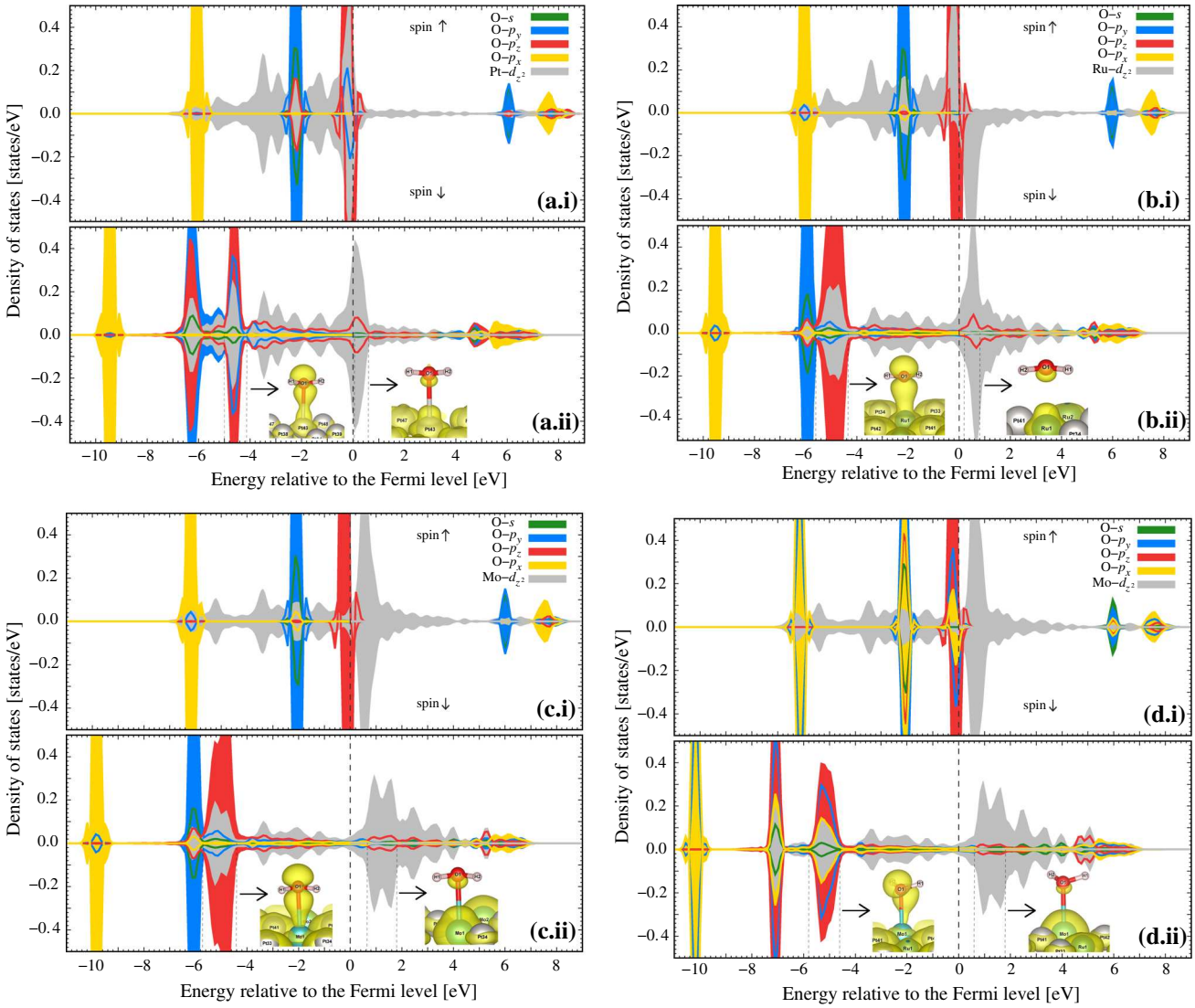


Fig. 2. (Above, labeled i) Partial density of states for some important orbitals of the water monomer before it interacts with metal surfaces of (a.i) Pt, (b.i) Pt-Ru, (c.i) Pt-Mo, and (d.i) Pt-Ru-Mo. (Below, labeled ii) Partial density of states for some important orbitals of the water monomer upon interaction with metal surfaces of (a.i) Pt, (b.i) Pt-Ru, (c.i) Pt-Mo, and (d.i) Pt-Ru-Mo. (Inset) Partial charge densities, which are plotted at a certain energy level indicated by the arrow

inset figures are assigned to related charge profiles. We took the DOS plot from the state soon before the H_2O interacts with surfaces rather than the gas phase, because when a molecule approaches the surface, its geometric structure changes. This naturally brought about changes in the electronic structure. It can be seen that the H_2O HOMO of Fig. 2(d.i), namely the $1b_1$ orbital, contains the most hybridized orbitals. The hybridization of s, p_x, p_y, and p_z orbitals of atomic O in H_2O can be seen most clearly here compared to the others. The hybridization of these molecular orbitals is also responsible for the interactions with surfaces. This is characterized by the disappearance of the peak of the molecular orbitals when it interacts with the surface. Moreover, new peaks are formed to the left and to the right of the H_2O HOMO. By further checking a charge profile plot at this energy level, which was between −5.8 eV and −4.6 eV, it has been proven that this is a bonding state. Thus, it can be said that the bonding state formed in 3 (d.ii) takes the most stable energy. Hence, interaction between a H_2O monomer and Pt-Ru-Mo surface is the most stable. On the other hand, the charge profile plot at the energy level of +0.5 to +2.0 eV, which is the new peak formed to the right of the H_2O HOMO, indicates an antibonding state. Thus, the antibonding states resulted in H_2O/Pt-Ru-Mo having the most unstable energy compared to the other three studied models. This reinforces the reason why H_2O adsorption on PtRuMo is the most stable compared to PtMo, PtRu, and Pt.

SUMMARY AND CONCLUSION

Ab initio DFT calculations were performed to study the adsorption of the water molecule on alloyed metal surfaces of Pt-Ru, Pt-Mo, and Pt-Ru-Mo. The H_2O molecule adsorbed on the surface of Pt-Ru and Pt-Mo with a parallel H_2O configuration with respect to the surface through the O atom. In contrast, on the surface of Pt-Ru-Mo, the H_2O adsorption is nearly parallel to the surface, i.e., the OH bond is tilted upward by 15°. These adsorption configurations determine the H_2O adsorption mechanism on metal surfaces.

The adsorption mechanism, which we analyze through the charge transfer that occurs during the adsorption, shows that the metal surface, which interacts directly with H_2O, releases electrons when it adsorbs the H_2O molecule. The Mo atom on the surface of Pt-Mo releases more electrons than Ru on Pt-Ru which releases more electrons than Pt on pure Pt(111) surfaces. This can be understood by the principle of the Pauling electronegativity difference for the chemical bonds. The electronegativity difference between H_2O and Mo, which is simply written as H_2O/Mo, is greater than H_2O/Ru which is greater than H_2O/Pt. Thus, the atom O with higher electronegativity would more readily attract electrons of Mo than electrons in Ru or Pt. This explains why the ionic character of covalent polar bonds in water/metal surfaces is highest in H_2O/Pt-Mo, followed by H_2O/Pt-Ru, followed by H_2O/Pt. Therefore, the interaction strength in H_2O/Pt-Mo is greater than H_2O/Pt-Ru which is greater than H_2O/Pt.

Furthermore, the interaction of H_2O with alloy metal surfaces was explained by the mechanism of natural chemical bond formation. The most inclined adsorption configuration of H_2O relative to the plane surface observed in H_2O/Pt-Ru-Mo leads to the charge transfer being less optimal. However, the water structure immediately before adsorption brings the most hybridized orbitals to the H_2O HOMO. The HOMO plays the most important role in the interaction, which is characterized by the disappearance of the HOMO and then forms the new states; bonding and antibonding states. In H_2O/Pt-Ru-Mo, the antibonding states are in the higher-energy position followed by Pt-Mo, Pt-Ru, and Pt. On the other hand, the bonding states of Pt-Ru-Mo take the most stable position, followed by Pt-Mo, Pt-Ru, and Pt. Thus, it is clear that the adsorption energy of a monomer H_2O/Pt-Ru-Mo > H_2O/Pt-Mo > H_2O/Pt-Ru > H_2O/Pt.

ACKNOWLEDGMENTS

We appreciate the support of the Directorate of Research and Community Services, Directorate General of Research and Development Strengthening, Ministry of Research, Technology and Higher Education of Indonesia through the grant of "Penelitian Dasar 2019" with the contract no. P/1805/UN23/14/PN/2019. WTC would like to thank Tondi, Clara, Ghozi, and Tina for supporting the H_2O data of the upright adsorption configuration system.

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